Interlinear Gloss as Chain-of-Thought Intermediate Step in Low-Resource Machine Translation: Hong Kong Hakka to Standard Written Chinese

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Introduction

Hong Kong Hakka (HK Hakka) is an endangered, low-resource Sinitic variety spoken in Hong Kong, and fieldwork observations indicate that most fluent speakers are over 65 years old. While a text-to-speech (TTS) system has been developed for HK Hakka, a publicly available machine translation (MT) system is still lacking. Previous studies have examined automated gloss generation (Okabe & Yvon, 2023; Ginn et al., 2024). Ramos et al. (2025) demonstrated that the effectiveness of gloss-assisted MT depends on the accuracy of model-generated glosses and noted that prompting is an alternative when linguistic annotations are insufficient for fine-tuning. Similarly, Ozer et al. (2026) found that glosses are helpful but sensitive to gloss quality. From the perspective of model training, Chung et al. (2024) showed that models with instruction fine-tuning that incorporated chain-of-thought (CoT) data outperform models without CoT data. Motivated by these findings, this study aims to develop an MT system for HK Hakka and investigates whether using interlinear gloss in CoT fine-tuning improves models' translation quality.

Methodology

We draw on a 561-pair working corpus that includes conversation, narratives, oral art forms, and oral history, and we annotate it with segmentation and interlinear glosses. We split the dataset into a train set of 479 pairs and a test set of 82 pairs, and fine-tuned Hy-MT2-7B and Qwen3.5-9B with QLoRA (Dettmers et al., 2023) and 3 training seeds. The models take Hakka as input and produce translations in standard written Chinese. The experiments have 3 conditions, namely (1) the base models, (2) the models fine-tuned with bilingual pairs (baseline), and (3) the CoT-fine-tuned models that produce Hakka segmentation, interlinear glosses of Hakka, and Chinese translation sequentially. For evaluation, we use BLEU and chrF2 as primary metrics, and BERTScore F1 as a supporting metric, while the intermediate segmentations and interlinear glosses are excluded from evaluation. We conduct paired bootstrap tests comparing the CoT condition against the baseline separately for each training seed, with 1000 resamples per test.

Results

Baseline Prompt

把下面的文本翻譯成繁體中文，不要額外解釋。

Sample User Input

<hak> 一毫子都冇分厓 </hak>

CoT Prompt

把下面的文本區隔，然後提供逐字註釋，
最後翻譯成繁體中文。

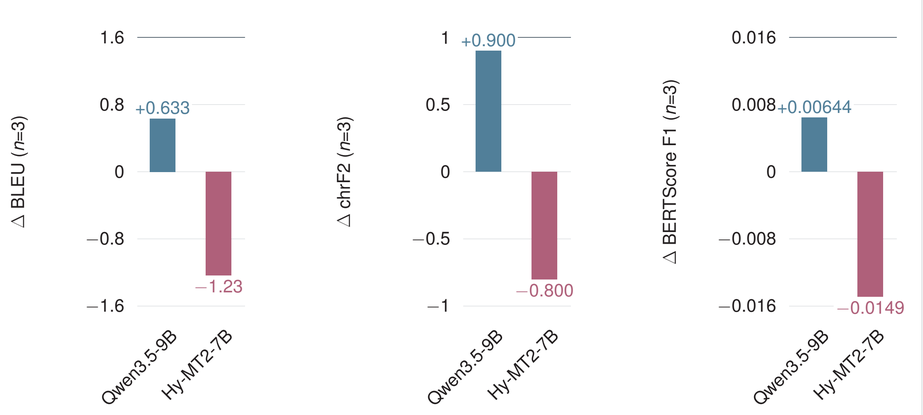
Sample Model Output

<hak-seg> 一 毫子 都 冇 分 厓 </hak-seg>
<gloss> one ten.cents also not.have give 1.SG </gloss>
<zh-Hant> 一毛錢也沒有給我 </zh-Hant>

Mean Translation Performance

Model	Metric	Base model (n = 1)	Baseline (n = 3)	CoT (n = 3)
Qwen3.5-9B	BLEU	16.6	52.2 (± 2.05)	52.8 (± 0.153)
	chrF2	14.9	43.6 (± 2.16)	44.5 (± 0.473)
	BERTScore F1	0.501	0.728 (± 0.0113)	0.734 (± 0.00588)
Hy-MT2-7B	BLEU	29.6	56.4 (± 0.635)	55.1 (± 1.78)
	chrF2	25.2	48.2 (± 0.493)	47.4 (± 1.92)
	BERTScore F1	0.561	0.751 (± 0.00578)	0.736 (± 0.00241)

**All values are reported to 3 significant figures. Baseline and CoT values are mean ± sample SD across training seeds.*



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The results indicate that the CoT condition outperforms the baseline for Qwen3.5-9B and underperforms the baseline for Hy-MT2-7B across the mean scores of BLEU, chrF2, and BERTScore F1. Both fine-tuning conditions also outperform their respective base models across all three evaluation metrics. However, only the CoT-baseline BLEU score difference at seed 42 for Hy-MT2-7B reaches statistical significance ($p = 0.047$, $\alpha = 0.05$).

Discussion

The effects of gloss in CoT vary across models and training seeds. Nevertheless, the CoT-fine-tuned models can generate segmentation and interlinear glosses in the expected format; hence, to our knowledge, they may serve as prototypes of the first automatic interlinear glossing system for HK Hakka. This highlights the potential of developing and deploying an automated gloss generation system for HK Hakka.

Conclusion

This study explored the effectiveness of gloss-based intermediate representations and found that CoT fine-tuning did not yield consistent improvement in this controlled experiment. This study also laid a foundation for the development of automated glossing systems for HK Hakka. Future work includes extending the experiments to more models, conducting in-depth qualitative analysis of the glosses and translations, and experimenting with alternative linguistic representations.

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github.com/hpsns414/hkhakka-mt